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RESEARCH ARTICLE

Assessing the climate change impacts and adaptation strategies for rice production in Punjab, Pakistan

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Abstract

Climate change is a serious threat to agriculture and food security. The development of adaptation measures is one of the key strategies to offset the climate change impacts on rice crop. A study was planned to develop adaptation strategies for future rice crop through simulation model. This study was consisted of six transplanting dates and three seedling densities which were tested to develop adaptations at three locations, i.e., Faisalabad, Hafizabad, and Gujranwala during the years 2015 and 2016. Climate change scenarios were developed for selected locations through statistical downscaling by selecting five general circulation models under representative concentration pathways (RCPs 8.5) for mid-century (2039–2069). The APSIM-ORYZA model was calibrated well with experimental data which showed an error of 2.35% between observed and simulated grain yield. The impact of climate change was studied by calculating the difference of baseline (1980–2010) and future yield. Model simulated results indicated that the rise in temperature will reduce the rice yield by 7.3%. To overcome this decrease in rice yield, suitable adaptation strategies were tested for mid-century. The developed adaptations, i.e., increased in plant population, nitrogen amount, and early transplanting, improved the rice yield by 8.7% under RCPs 8.5 for mid-century scenarios. Overall, this study provide better understanding of the adaptation processes for sustainable rice yield under anticipated future climate change.

Keywords Climate change · APSIM-ORYZA · Climate impacts · Adaptations

Introduction

Climate change has significant impacts on agriculture and food security (Schlenker and Lobell 2010; Downing 2013; Pachauri et al. 2014; Alexeeff et al. 2018). Climate projections indicate that global temperature is expected to rise by 2.5 °C up to 2050, which will reduce the crop production (IPCC 2013a). There is a dire need to assess the impact of climate change and to develop adaptation strategies to reduce the severity of climate change on crop production.

Rice is a staple food for more than half of the world's population (FAO 2013). In Pakistan, rice is the second major crop after wheat, which is cultivated on an area of 2.8 million hectare with production of 7.2 million tons (Government of Pakistan 2019). However, rice yield is less than the potential due to various factors such as high temperatures and low and high erratic rainfall (Siddiqui et al. 2012; Ahmad et al. 2015a; Shakoor et al. 2015). It is worth mentioning that not all the impacts of climate change are negative, but there is a trade-off between crop production gains and losses. On one side, high temperature and low rainfall may affect the yield negatively, while on the other side, increased CO_2 concentration may enhance photosynthesis rate. However, the prominent climate change impacts on rice yield are negative (Wassmann et al. 2009b; Liersch et al. 2013; Soora et al. 2013). Ahmad et al. (2015a) reported that rice yield will decline by 17% due to 2.8 °C rise in temperature of Pakistan under RCP 8.5 for mid-century scenario (2039–2069). Every 1 °C increase in minimum temperature during the growing period decreased the rice yield by 10% at International Rice Research Institute (Peng et al. 2004; Shah et al. 2011). During the growing season, higher temperature reduced the grain yield due to shorter growing period (Jagadish et al. 2015), less light interception, and poor kernel weight gain (Jagadish et al. 2007; Wu et al.

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2016). Anthesis stage of rice was very sensitive to a temperature above 38 °C even if exposed for a duration of an hour (Wu et al. 2016). Although the future elevated CO_2 concentration has positive effect on growth, but the associated rise in temperature will nullify this favor to growth and yield of rice (Kimball et al. 2002; Krishnan et al. 2007).

The impact of climate change on rice yield has been well documented (Nakagawa et al. 2003; Kang et al. 2009; Auffhammer et al. 2012; Kim et al. 2013; Ahmad et al. 2020). However, adaptations are the key strategies to mitigate the negative impacts of climate change on agricultural crops like rice (Füssel 2007; Venkateswarlu and Shanker 2009; Wassmann et al. 2009a; Biesbroek et al. 2010). The adaptations include the changes in management practices such as adjustment of transplanting dates (Liu et al. 2018), plant population (Ahmad et al. 2015a), and optimization of nutrients (Dobermann et al. 2002). Results of different studies indicated that such adaptations are cultivar and region specific (Smith and Olesen 2010). Thus, selection of suitable adaptation strategy is important to reduce the climate change impacts in future (Howden et al. 2007).

Crop growth simulation models are valuable tools and have been used to evaluate the impact of climate change on field crops (Lobell et al. 2008; White et al. 2011; Asseng et al. 2015; Rahman et al. 2018; Ullah et al. 2019; Ahmad et al. 2020). The Agriculture Production System sIMulator (APSIM) model has been widely tested and used in climate change impact assessments (Keating et al. 2003; Hammer et al. 2010; Gaydon et al. 2017). For rice crop, the "ORYZA" module was developed by incorporating the ORYZA2000 rice growth model to the APSIM modeling framework (Gaydon et al. 2012a; Gaydon et al. 2012b), which has been successfully used to assess the impact of climatic vulnerabilities on rice (Liu et al. 2013). The climatic risks and vulnerabilities could accurately be assessed through combine use of crop and general circulation models (GCMs) (Stevens and Madani 2016). GCMs are used to predict the future climatic conditions (IPCC 2013b). The statistical downscaling of GCMs provides the high-resolution and site-specific climatic data for impact analyses (Ayar et al. 2016).

Studies have shown that various techniques were used for assessment of climate change on rice. For example, Fang et al. (2004) used statistical regression, while process-based model were used by Tao et al. (2008), Liu et al. (2013), Ahmad et al. (2015a), and Bai et al. (2016). These studies were region specific, but none of these studies developed climate change adaptations using high-resolution climatic data with a crop growth model in different environments. However, current study was planned with the objective to develop climate change adaptation strategies for mid-century (2039–2069) scenarios for rice in Punjab, Pakistan, using APSIM-ORYZA modeling approach.

Materials and methods

Field experiment

Field experiments were conducted at three locations, i.e., Faisalabad, Gujranwala, and Hafizabad during the years 2015 and 2016. Experiments were consisted of six transplanting dates (10 June, 20 June, 30 June, 10 July, 20 July, and 30 July) and three seedling densities (1 seedling/hill, 2 seedlings/hill, and 3 seedlings/hill). The rice nursery was sown 1 month before transplantation on beds using seed rate of 30 g per m^2. Seedlings were transplanted manually in the puddled field after 30 days. The recommended doses of 150 kg ha^{-1} nitrogen (N) in the form of urea, 100 kg ha^{-1} phosphorus (P) in the form of diammonium phosphate, and 75 kg ha^{-1} potash (K_2O_5) in the form of sulfate of potash were applied. All P, K_2O_5, and one third N fertilizers were applied at the time of transplanting, while the remaining dose of nitrogen in two splits at 15 and 30 days interval after 1st dose.

Collection of experimental data for model

Crop phenological data (days to anthesis and maturity) were recorded by tagging ten plants from the middle of row in each experiment. Crop growth data including leaf area index (LAI) and total dry matter (TDM) were collected fortnightly. At each fortnight, for LAI and TDM, one foot area was harvested, fresh weight was recorded, and then 10-g subsample of leaves was taken to record the LAI using LI-3000 area meter (Lincoln Nebraska, USA). The assessed leaf area of 10-g subsample was converted to total crop leaf area through unit method. TDM was calculated by oven drying the subsample at 70 °C for 48 h. Similar method of LAI and TDM calculation was used by Ahmed et al. (2018). The data of grain and biological yield were recorded by harvesting the crop at maturity. The crop management data such as transplanting dates, planting methods, fertilizers, irrigation amounts, and irrigation dates were used to create input files for model.

Soil and weather data for model

Soil series data of each location were collected from the Soil Survey of Pakistan (SSOP). The Lyallpur soil series was used for Faisalabad, Jaranwala series for Gujranwala, and Hafizabad soil series for Hafizabad districts. The soil profile data of silt%, clay%, organic carbon%, pH in water, cation exchange capacity (cmol/kg), and total nitrogen% were collected and used as input for model. The organic carbon (OC%) was calculated by dividing the organic matter by 1.70 (Bowman 1997). The parameters such as permanent wilting point%, field capacity%, and bulk density (gcm^{-3}) were calculated by the model using the methods describes by Rawls et al., (1982). According to USDA classification, the Lyallpur

and Jaranwala series were coarse silty, while Hafizabad series was fine loam. The less organic carbon was found in the lower soil profile too. In Faisalabad series, soil samples taken to a depth of 150-cm maximum and minimum silt, clay, organic carbon, and soil pH were 53–58%, 10–17%, 0.05–0.53%, and 8.28–8.37, respectively. In Hafizabad series, cm maximum and minimum silt, clay, organic carbon, and soil pH were 48–53%, 15–18%, 0.03–0.52%, and 8.00–8.2, respectively, while the maximum and minimum silt, clay, organic carbon, and soil pH in Jaranwala series were 56–65%, 12–14%, 0.04–0.59%, and 8.2–8.3, respectively (Table 1).

Daily weather data of maximum temperatures (T_{max}), minimum temperatures (T_{min}), rainfall (mm), wind speed (km h^{-1}), and sunshine hours were collected from the weather observatory of Pakistan Meteorological Department (PMD) for the year 2015 and 2016 as shown in Fig. 1. The solar radiations for model input were calculated by using Angstrom formula (Angstrom 1924). The temperatures of about 1 °C was warmer during the season 2016 as compared with 2015, However higher rainfall was recorded during the month of August in 2016 (Fig. 1). High rainfall of 241 mm at Hafizabad and 323 mm at Gujranwala was recorded during the season 2016 as compared with 2015.

Model calibration and validation

The ORYZA2000 model under $APSIM_{V7.7}$ was used for this study. APSIM-ORYZA is a dynamic eco-physiological crop model which simulates the rice phenology, growth, and yield in response to soil, weather, crop management, and plant genetics (Bouman and Van Laar 2006; Holzworth et al. 2014). Genetic coefficients of model were adjusted to the local conditions under non-stressed treatment. Model was calibrated with transplanting date of 30 June and 2 seedlings per hill at three locations during the year 2016. After calibration, the model was run with remaining transplanting dates and number of seedling per hill for all locations during 2016 and the data for the year 2015. Performance of model and reliability of estimated genetic coefficients was assessed by calculating different statistical indices such as root mean square error (RMSE), error percentage, and Willmott concordance index (d). The root mean square error (RMSE) in Eq. 1 describes the size of error produced from model. Percent error in Eq. 2 shows the error between observed and simulated values, while enhanced Willmott concordance index (d) describes the deviation between observed and predicted values.

$$RMSE = \left[\sum_{i=1}^{n}(Pi-Oi)^2/n\right]^{0.5} \quad (1)$$

$$Error\ (\%) = \left(\frac{Oi-Pi}{Oi}\right) 100 \quad (2)$$

$$d = 1-\left[\frac{\sum_{i=1}^{n}(Pi-Oi)^2}{\sum_{i=1}^{n}(|Pi|+|Oi|)^2}\right] \quad (3)$$

where n is the number of variables and I is the ith quantity of predicted (p) and observed (O) values.

RMSE represents the error between treatments. It clarifies the uncertainty in the data. If $d = 1$, it means that the model performance is perfect. If $d = 0$, it indicates a model for which the mean square error is just matching the variability in the observed data. Negative value of d indicates that the average observed values are better than the model values in any case (Krause et al. 2005; Gaydon et al. 2017).

Statistical downscaling and climate change projections

Baseline climate data (1980–2010) on T_{max}, T_{min}, solar radiations, rainfall, and wind speed were collected from the Pakistan Meteorological Department (PMD). The future climate change scenarios were generated for mid-century (2039–2069) under Representative Concentration Pathway 8.5 (In RCP 8.5, it is assumed that global annual greenhouse gas emissions continue to rise throughout the twenty-first century) using the delta method approach (Wilby et al. 2004). General circulation models (GCMs) also known as the numerical models are the climate models representing the physical processes in the atmosphere, ocean, cryosphere, and land surface. They are basically used for simulating the climate systems in response to climate change factors, e.g., greenhouse gases. The five GCMs were selected to represent the uncertainty in prediction of temperatures and precipitation for all locations

Table 1 Soil series data used as input for model at each location

Locations	Faisalabad				Hafizabad				Gujranwala			
Depth (cm)	Silt%	Clay%	OC%	pH	Silt%	Clay%	OC%	pH	Silt%	Clay%	OC%	pH
0–9	56	10	0.53	8.28	48	15	0.52	8.2	57	14	0.59	8.2
9–29	53	13	0.20	8.35	49	17	0.24	8.2	56	14	0.57	8.2
29–98	53	17	0.13	8.25	49	18	0.12	8.1	56	13	0.07	8.2
98–133	54	16	0.11	8.32	53	18	0.07	8.1	62	12	0.05	8.2
133–150	58	12	0.05	8.37	52	18	0.03	8.0	65	14	0.04	8.3

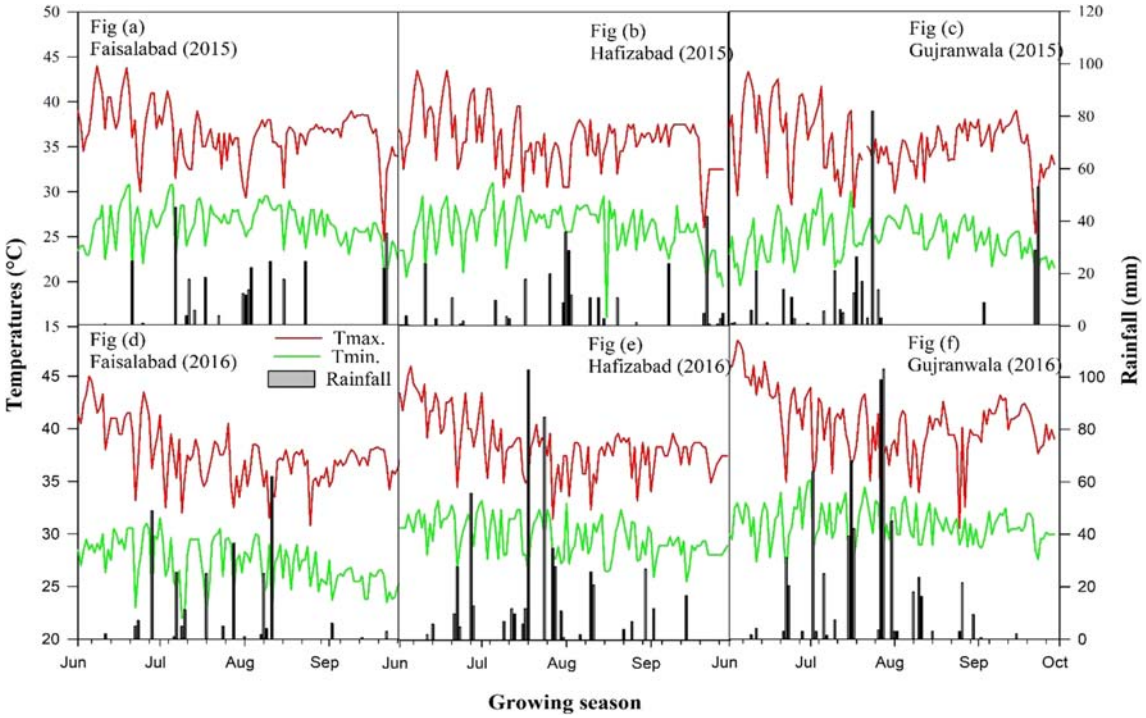


Fig. 1 Seasonal weather data of maximum, minimum temperature, and rainfall at Faisalabad, Hafizabad, and Gujranwala during the years 2015 and 2016

from the Coupled Model Inter-comparison Project (CMIP5) (Taylor et al. 2012). Mean monthly changes in mean T_{max}, T_{min}, and precipitation were calculated and compared with future 30-year climate to baseline; after that, monthly changes in the standard deviation (SD) of T_{max}, SD of T_{min}, and rainy days (> 0.1 mm) were calculated and compared with the future climate to baseline. In GCMs simulations, the shape parameters of the gamma distribution for wet event is not the sufficient quality; therefore, mean monthly changes were imposed to baseline climate using stretch distribution approach (Ruane et al. 2015). During this process, the solar radiations, wind speed, and humidity were assumed unchanged. Finally, the climate scenarios were developed for the cool/wet, cool/dry, hot/wet, hot/dry, and middle GCMs. The detail protocol is described by Rosenzweig et al. (2015).

Impact of climate change and development of adaptation strategies for rice

The calibrated APSIM-ORYZA model was used to evaluate the impact of climate change on rice for mid-century scenarios. Seasonal files were created from the management practices of optimum sowing date of 30 June with 2 seedlings per hill for all locations. The baseline climate and future scenarios were used to create the weather files for each location. The CO_2 concentration of 571 ppm was used for RCP 8.5 under mid-century, proposed by IPCC (2014). The impact of climate change was calculated from future and baseline mean yield using Eq. 4. The climate change adaptations strategies were developed by modifications in crop management practices in the model. The selected adaptation strategies were increasing planting density, amount of nitrogen, change in planting time, development of heat tolerant cultivars, and decreasing amount of irrigation. The model was tested against the set of different values/practices of these adaptation strategies. The values showing the maximum increase in yield under mid-century scenario for all the locations were considered as the adaptation strategies for rice. The detail flow diagram for climate change analysis and development of adaptation strategies is shown in Fig. 2.

$$\%Change = \left(\frac{Simulated - Observed}{Observed}\right) \times 100 \quad (4)$$

Results

Calibration and validation of APSIM-ORYZA model

The genetic coefficients of phenology, growth, and yield for rice cultivar Basmati-515 estimated by using APSIM-ORYZA at three locations are presented in Table 2. The model-estimated growing degree days for vegetative and flowering stage were 0.000520 and 0.00934, respectively. The panicle development stage took more days (Table 2).

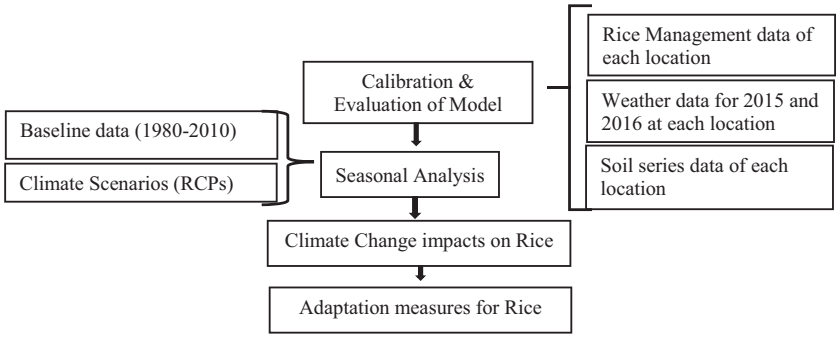


Fig. 2 Flow diagram on development of adaptation strategies under climate change for rice

The maximum photoperiod of 12.36 h was recorded which implies that day length higher than 12.36 h would reduce the development rate of rice.

Calibration results of APSIM-ORYZA showed a close agreement between observed and simulated values for phenology, biomass production, and yield as shown in Table 3. Model predicted the phenology (days to anthesis and maturity) with 1-day difference from observed values. Good prediction in phenology could be due to the presence of soil moisture and good germination. Model simulated the maximum LAI well with percent error of 3.06%. A good agreement was found between observed and simulated grain yield with percent error of 2.24%. Biomass was also well simulated by the model with 4.27% error (Table 3). Thus, results revealed that model was calibrated well with percent error in the range of 0.90 to 3.06%.

Model was run with the independent field experiment data, i.e., remaining transplanting dates and number of seedling per hill of 2015 and 2016 to validate the performance of model. The results indicated that model showed close association between observed and simulated days to anthesis and maturity in both years (Table 4). RMSE values showed a 2–3 days difference between observed and simulated days to anthesis and maturity (Table 4). Model underestimated the phenological events for early transplanting (10–20 June and 20–30 July) with 3 seedlings per hill. Low and high temperatures in early and late transplanting with high transplanting density could reduce the growing season length of plant. The model performance was good in simulation of grain yield in 2015 and 2016, which showed a % error ranged from 2.24 to 15.6% and RMSE values of 482 kg ha^{-1} in 2016 and 520 kg ha^{-1} in 2015. Model underestimated the yield in early and late transplanting with 3 seedlings per hill. The plant competition for nutrient and light due to more plant population resulted a decline in yield. The performance of the model in simulation of biomass was satisfactory during both years with RMSE values of 681 kg ha^{-1} in 2015 and 773 kg ha^{-1} in 2016. Higher error percent was found in years 2015 as compared with 2016. Model underestimated the biomass in early and late transplanting with 1 seedling per hill. The overall results of model validation indicated its performance as acceptable because all the RMSE values were within the range of observed variability (observed standard deviation) of the experiment (Table 4).

Table 2 Adjusted genetic coefficient of APSIM-ORYZA for rice cultivar Basmati-515

Cultivar	DVRJ[a]	DVRI[b]	DVRP[c]	DVRR[d]	MOPP[e]	PPSE[f]
Basmati-515	0.000520	0.00934	0.001161	0.00232	12.36	0.395

DVRJ[a] is the development rate in juvenile phase (°Cd-1)
DVRI[b] is the development rate in photoperiod sensitive phase (°Cd-1)
DVRP[c] is the development rate in panicle development (°Cd-1)
DVRR[d] is the development rate in reproductive phase (°Cd-1)
MOPP[e] is the maximum optimum photoperiod (h)
PPSE[f] is the photoperiod sensitivity (h-1)

Climate change projections

Projected climate results of five GCMs indicated an increasing trend for T_{max}, T_{min}, and precipitation as compared with the baseline as shown in Table 5. Mean temperatures of 2.5 °C at Faisalabad, 2.9 °C at Gujranwala, and 3.0 °C at Hafizabad were projected to rise under RCP 8.5 for mid-century scenarios. Greater increase in temperature was found in hot/dry GCMs, while middle and cool/wet GCMs indicated less increase in temperature as compared with other GCMs. The annual rainfall was also projected to increase by 104 mm at Faisalabad, 319 mm at Gujranwala, and 270 mm at Hafizabad. The cool/dry and hot/dry GCMs indicated a decreasing trend in precipitation at Faisalabad, while increasing trend at Gujranwala and Hafizabad. More increase in precipitation

Table 3 Calibration of APSIM-ORYZA at all locations using treatment 30 June with 2 seedlings per hill during the year 2016

Parameters	Observed	Simulated	% Error
Days to anthesis (days)	77	78	1.30
Days to maturity (days)	111	112	0.90
Leaf area index maximum	5.89	6.07	3.06
Grain yield (kg ha^{-1})	3933	4021	2.24
Biomass (kg ha^{-1})	13,691	14,275	4.27

Table 4 Evaluation of APSIM-ORYZA showing % error for phenology, biomass, and yield for all locations during the year 2015 and 2016

Sowing dates	Seedling per hills	Days to anthesis		Days to maturity		Kernel yield		Biological yield	
		2015	2016	2015	2016	2015	2016	2015	2016
10 June	1	5.10	5.56	3.07	3.17	12.39	11.86	15.58	− 10.62
	2	4.44	6.67	4.07	4.76	13.96	10.14	16.54	15.44
	3	− 4.44	6.59	− 3.10	3.15	13.15	− 12.38	19.65	11.85
20 June	1	4.88	4.76	3.54	3.39	7.40	3.91	8.17	7.58
	2	4.88	4.71	1.63	1.69	4.38	3.98	9.71	8.89
	3	− 2.44	2.30	− 1.75	2.50	7.19	− 6.82	11.62	11.7
30 June	1	4.25	5.10	3.85	4.05	15.0	6.54	15.84	15.47
	2	1.12	1.30	1.21	0.90	2.24	2.30	4.27	5.10
	3	3.55	2.55	5.68	1.88	10.87	9.88	10.44	11.08
10 July	1	2.74	2.70	0.95	2.68	9.10	7.85	9.09	− 6.97
	2	2.7	2.67	0.94	2.65	7.36	4.34	6.53	5.44
	3	2.7	1.33	1.83	2.60	8.59	5.36	10.32	5.05
20 July	1	2.86	2.78	2.97	1.90	15.61	13.24	10.27	8.21
	2	2.82	1.39	1.98	− 0.95	− 9.76	7.03	12.66	8.24
	3	− 1.39	1.39	− 2.94	1.89	10.07	9.89	− 14.2	9.87
30 July	1	1.54	2.9	2.22	2.11	6.26	5.71	17.7	− 15.14
	2	1.52	2.86	2.2	0.00	9.03	8.23	16.39	13.54
	3	− 1.52	− 2.82	4.35	− 1.89	− 6.64	9.81	18.63	16.87
$RMSE^{average}$		2.93	3.16	2.61	2.72	520	482	681	773
$SD^{observed}$		5.75	6.04	8.04	9.52	523	514	1000	1113

$RMSE^{average}$ is the average root mean square error, $SD^{observed}$ is the standard deviation of observed data

was found in cool/wet GCMs as compared with other GCMs. While from ensemble mean of all GCMs, it can be concluded that T_{max} of 3.07 °C and T_{min} of 3 °C may increase, while rainfall may increase from 104 mm to 319 mm in rice zone of Punjab Pakistan under RCP 8.5 for mid-century scenarios.

Impact of climate change on rice yield under RCP 8.5 for mid-century (2039–2069)

In the current study, climate change was projected to decrease rice yield at three locations due to rise in temperature and variation in precipitation (Table 6). Climate change results indicated that on average rice yield will decline by 8.6% at Faisalabad, 7.1% at Hafizabad, and 6.4% at Gujranwala. In case of hot/dry GCMs, a greater reduction in rice yield, 15% was recorded (Fig. 3a–c). The reduction in mean yield, 1.5% in middle GCMs and 3.8% in cool/wet, was observed. However, higher reduction in yield, 4.9% was observed in hot/dry GCMs and 11.5% in hot/wet. The variation in yield in all GCMs at three locations is presented in Fig. 3a–c. It was concluded that mean yield will be reduced by 7.3% in rice zone under RCP 8.5 for mid-century scenarios.

Table 5 Change in temperatures and rainfall in all GCMs under RCP 8.5 for mid-century (2039–2069) at three locations

Locations/ GCMs	Faisalabad			Hafizabad			Gujranwala		
	ΔT_{max} (°C)	ΔT_{min} (°C)	ΔRainfall (mm)	ΔT_{max} (°C)	ΔT_{min} (°C)	ΔRainfall (mm)	ΔT_{max} (°C)	ΔT_{min} (°C)	ΔRainfall (mm)
Middle	1.21	0.97	196	2.00	1.77	302	1.88	1.53	311
Cool/wet	1.98	1.50	232	2.51	2.21	333	2.41	1.98	347
Hot/wet	3.07	3.97	182	3.43	3.87	313	3.39	3.58	346
Cool/dry	2.20	1.88	− 54	2.87	2.42	216	2.83	2.20	393
Hot/dry	3.79	4.03	− 36	4.55	4.76	190	4.58	4.78	198
Average	2.45	2.47	104	3.07	3.00	270.8	3.03	2.81	319

Table 6 Percent decrease in rice yield transplanted on 30 June with 2 seedlings per hill under RCP 8.5 for mid-century

GCMs	Faisalabad	Hafizabad	Gujranwala
Middle	− 2.00	− 1.55	− 0.93
Cool/wet	− 4.49	− 3.86	− 3.12
Cool/dry	− 6.55	− 4.29	− 4.02
Hot/wet	− 13.31	− 11.50	− 9.90
Hot/dry	− 16.43	− 14.34	− 14.14

Impact of climate change adaptation strategies on rice yield

Climate change adaptation strategies were developed to reduce the impact of climate change. Various management options were tested in the model for the development of adaptation strategies, and the most suitable set of options were selected as adaptation, which are shown in Table 7. Firstly, these management practices were checked one by one by altering them in the model management portfolio and looking the model output. Secondly, a set of management options were tested against the mid-century scenario predicted climate. The best set of these options in which the model produced maximum yield on experimental locations was selected as the suitable adaptation strategies for mid-century. The results indicated that by modification in production technology of rice such as increase in planting density, nitrogen amount, decrease in irrigation, and early transplanting of rice will bring increase in yield.

Further results showed that by application of developed adaptation strategies, the rice yield will be increased by 7.3% at Faisalabad, 9% at Hafizabad, and 9.8% at Gujranwala. In case of GCMs, the middle and cool/wet GCMs showed an increased in yield of about 14%, while cool/dry GCMs indicated an increase in yield by 9.8% and hot/wet by 5.3%. The hot/dry GCMs did not show any increased in yield (Fig. 4a–c). From the above results, it was concluded that rice yield will increase by 8.7% due to the adaptation strategies for changing climate of mid-century scenarios.

Discussions

The APSIM-ORYZA model was calibrated and validated using field experiment data. The genetic coefficients for phenology, growth, and yield were estimated according to field conditions. The cardinal temperatures for rice were fixed in the model; thus, photoperiod sensitivity parameters were estimated. Six coefficients presented in Table 2 indicate the genetics and physiology of plant was initially estimated under controlled experiments. But in the absence of such experiments, these coefficients were estimated by fitting the model with field experiment data. The coefficients were estimated through repeated iteration until a close match between observed and simulated values was achieved. The values of estimated coefficients in the current study were within the range estimated by Nissanka et al. (2015); however, value of maximum optimum photoperiod in current study was greater by 2 h. The reason behind which was the season specificity of each cultivar.

Calibration results showed that model was well set to observed data. The close agreement was found between observe and simulated days to anthesis and maturity (Table 3). The precise estimation of phenological values captures the genotypic variation such as leaf development, dry matter production, and grain yield (Robertson et al. 2002). The close fit was recorded in simulation of biomass and yield also (Table 3).

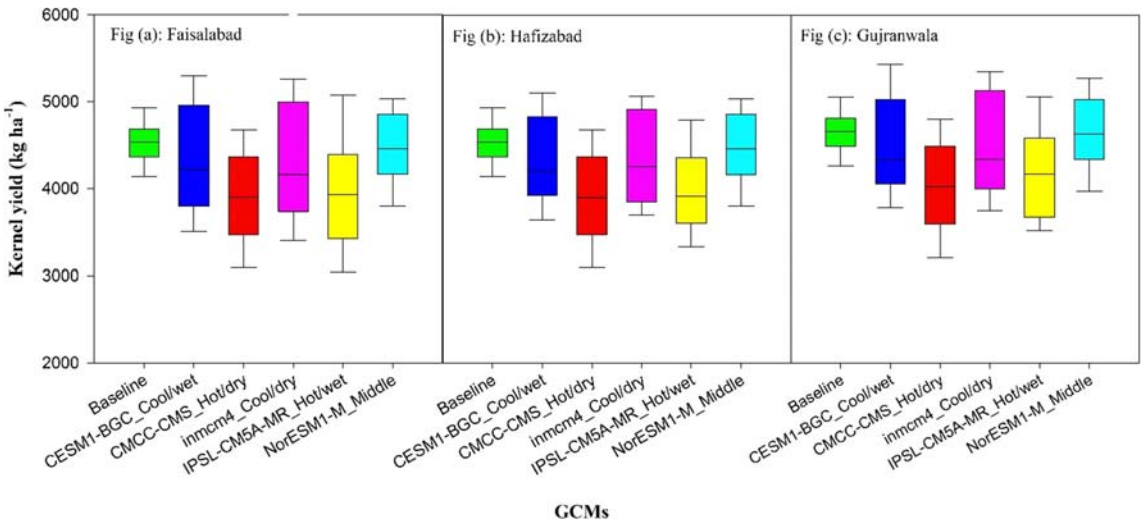


Fig. 3 Impact of climate change on rice yield at three locations under RCP 8.5 for mid-century scenarios under current management

Table 7 Developed adaptation package for rice under RCP 8.5 for mid-century scenarios

Sr. No.	Variables	% Change
1	Increase in nitrogen amount (kg ha^{-1})	12
2	Increase in plant density (m^{-2})	20
3	Decrease in number of irrigation	10
4	Change in transplanting date	8 days earlier
5	Developed heat tolerant cultivars up to 3 °C	Genetic coefficients

Less error in simulation of biomass and kernel yield by APSIM-ORYZA was also reported by Belder et al. (2007) and Fang et al. (2004). Less error of the model simulated data as compared with the standard deviation of observed experimental data indicate that model predicted the variable of interest within the bound of experimental uncertainty and closest to reality (Gaydon et al. 2017). The validation results of the model showed minimum error and RMSE for phenology and yield attributes of rice. The model underestimated the phenology in early and late transplanting with 3 seedlings per hill (Table 4). It could be due to low and high temperature that caused the poor germination, growth, and development which resulted decrease in maturity and anthesis days. Similar was found that high temperature reduced the crop duration; the rise in temperature by 2 °C reduced 6 days in maturity of rice (Rani and Maragatham 2013). The model underestimated the biomass with 1 seedling per hill in early and late transplanting (Table 4). The low and high temperatures reduced the biomass production in rice due to physiological changes (Ohsumi et al. 2012).

Downscaling is the process to take information known at large scale to make predictions at local scale. Statistical downscaling is a two-step process. The development of statistical relationships between local climate variables and application of these relationships to the output of general circulation models to simulate local climate characteristics in future (Flower et al. 2007). Delta scenarios (Wilby et al. 2004) are based on historical baseline daily weather data in which weather variables of each day are perturbed by following the changes in climate model outputs mid-century RCP 8.5 against those same model output for historical time period. The selection criteria of GCMs were their better performance in the region and reputation of the developing institute (Rosenzweig et al. 2014). Future projections showed that temperature is expected to rise in rice zone of Pakistan under all GCMs for RCP 8.5 for mid-century. There was uncertainty in prediction of precipitation (Table 5). It has been reported that temperatures are expected to rise in Pakistan due to deforestation and emissions of greenhouses gases (Ahmad et al. 2015b). The climate analyses showed that the future rise in temperature will result in erratic, intense rainfalls, late monsoon seasons, and dry winter (Ahmad and Rasul 2018).

Climate change has negative effect on rice yield. The current study found that rice yield would be reduced by an average of 7.3% under RCP 8.5 for mid-century across all GCMs (Table 6). The decline in yield was due to projected rise in temperature as shown in Table 5. High temperature associated decrease in yield was primarily due to reduction in the

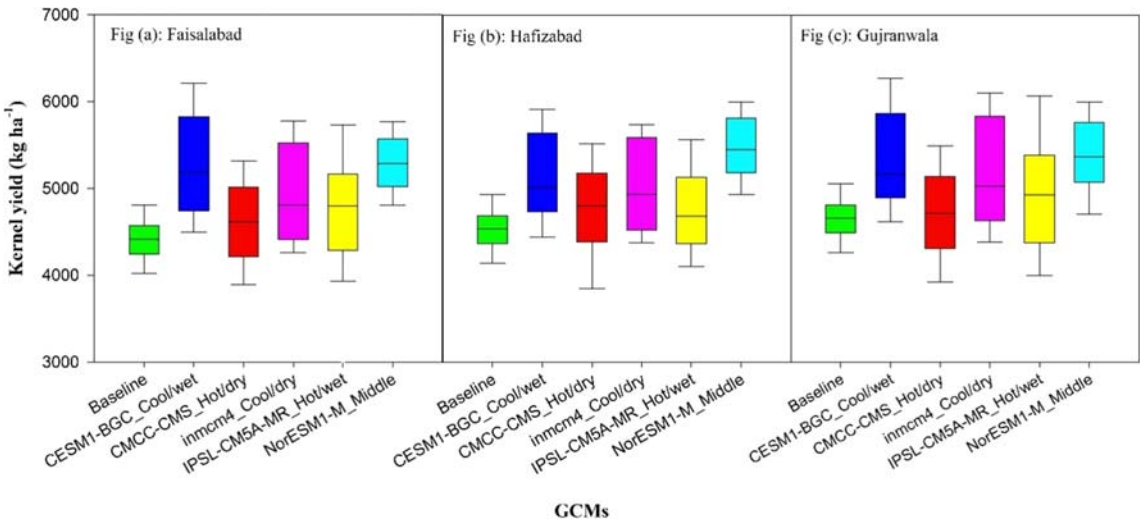


Fig. 4 Impact of climate change adaptations on rice yield at different location under RCP 8.5 for mid-century scenarios

growing period. The average daily temperature for rice growth is 20–25 °C. Increase in temperature above optimum reduced the grain size and weight that resulted in lower yield. High temperature also reduced the grain filling duration that limited the time for grain development, and yield was declined (Krishnan et al. 2011).

The developed adaptation strategies reduced the impact of climate change. The current study results showed that yield would be increased by 8.7% due to adaptation strategies (Fig. 4). It was reported by Khanal et al. (2018) that adaptations can reduce the climate induced yield losses. The modifications in production technology of rice such as increase in plant population, nitrogen amount, and early transplanting could help to reduce the impact of climate change (Table 7). Farmers are also using less amount of nutrients then recommended (Khaliq et al. 2017). Increasing the amount of nitrogen may contribute to better early crop establishment and increased biomass accumulation. Furthermore, high nitrogen means more chlorophyll and subsequently more food preparation in a given period of time. The plant population in the farmer's field is generally less than the recommended; thus, increasing number of plants could utilize the resources such as water, nutrient, and light efficiently. The development of climate resilient cultivars of rice through biotechnological strategies can aid to overcome the impact of climate change. The decrease in amount of the irrigation is associated with the development of heat tolerant, dwarf, improved harvest index, and low seasonal transpiration losses cultivars. Transplanting rice earlier escapes it from high temperature at anthesis stage which is one of the major causes of low rice yield due to increased sterility. Therefore, the development of adaptations through modifications in management of rice crop and development of heat-tolerant cultivars are useful strategies to offset the impact of climate change.

Conclusion

The crop model APSIM-ORYZA was calibrated and validated with 2 years field experiment data, which showed a good agreement between observed and simulated phenology, growth, and yield. The % error ranged from 0.95 to 16.3%. Future projection showed that T_{max} of 2.8 °C and T_{min} of 2.7 °C and rainfall of 231 mm would increase in rice zone (Gujranwala, Hafizabad and Faisalabad) of Punjab, Pakistan. Future rise in temperature will reduce the yield by 7.3% in the discussed rice zone. The developed adaptation strategies in the form of increased plant population, nitrogen amount, and early transplanting improved the rice yield by 8.7% under RCP 8.5 for mid-century scenarios. Farmers, agronomist, breeders, and stakeholders will mainly get the benefit of this study.

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